

CS 307: Software Engineering

Homework #2

Due on March 5th, 2026 at 11:58pm

Team #26: BoilerBridge

Team Members: Logan, Shreyansh, Xavion Marable, Emilio Guzman,

Problem 1

This is a group homework problem. Your group will be the same as the one that you are enrolled in for the team project in this course.

Find one open-source project that is closest to your team project. Reverse engineer the design and a few architectural/design patterns from that project.

Describe the above reverse engineered details in the following format. If applicable, provide pointers to project's directories/files/classes in your descriptions and/or diagrams.

1. Purpose (1 point)

a. Briefly explain the chosen open-source project and its purpose.

This is an AI powered travel planning application that is designed to help users organize and manage trips more easily. This is a full-stack, React Native application that uses generative AI to create personalized travel recommendations and assist users in planning itineraries. This application automatically suggests destinations, activities and accommodations based on user preferences using AI. Its purpose is to let users log in, create trips/itineraries, get AI-generated travel plans (Gemini) and use Google locations services while storing trip and user data in Firebase.

b. Provide the link to the open-source project repository.

<https://github.com/zahraaMeky/ai-travel-planner-app>

2. Design Outline (2 points)

a. Identify its components and describe the purpose of each component.

There are 8 main components for this React-Native application, with each one of them having subcomponents.

1. **app/** → Holds the main application screens and routing logic that controls navigation between features like authentication, trip creation, trip details and the main tab interface
2. **components/** → Holds reusable UI components and feature modules like login and trip creation that are used across multiple screen in the app.
3. **config/** → Stores the configuration files for external services

like Gemini and Firebase. Enables authentication, database access and AI generated travel recommendations

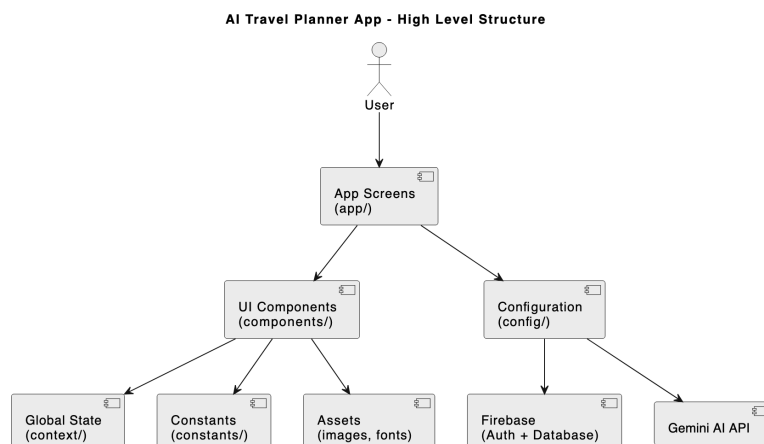
4. **context/** → Implements React Context to manage and share global state, like trip creation data, across different parts of the application.
5. **constants/** → Holds shared constants like colors and predefined data to ensure a consistent style and reusable values in the app.
6. **assets/** → Contains static resources like images and fonts that are used by the UI.
7. **scripts/** → Holds utility scripts used for development tasks like resetting or maintaining the project.
8. **Root Configuration Files** → Holds files like package.json, app.json that manage project dependencies, application settings and build configurations.

b. Describe the interactions between components.

The **app/** component is the central controller of the app. It loads the different screens and uses the **components/** folder to render the user interface elements for login, trip creating, viewing trip details, etc. When creating a trip or logging in, the screens call services that are defined in the **config/** directory. The configuration files connect the app to external services like Firebase for auth and database storage, and the Gemini AI for generating travel itineraries.

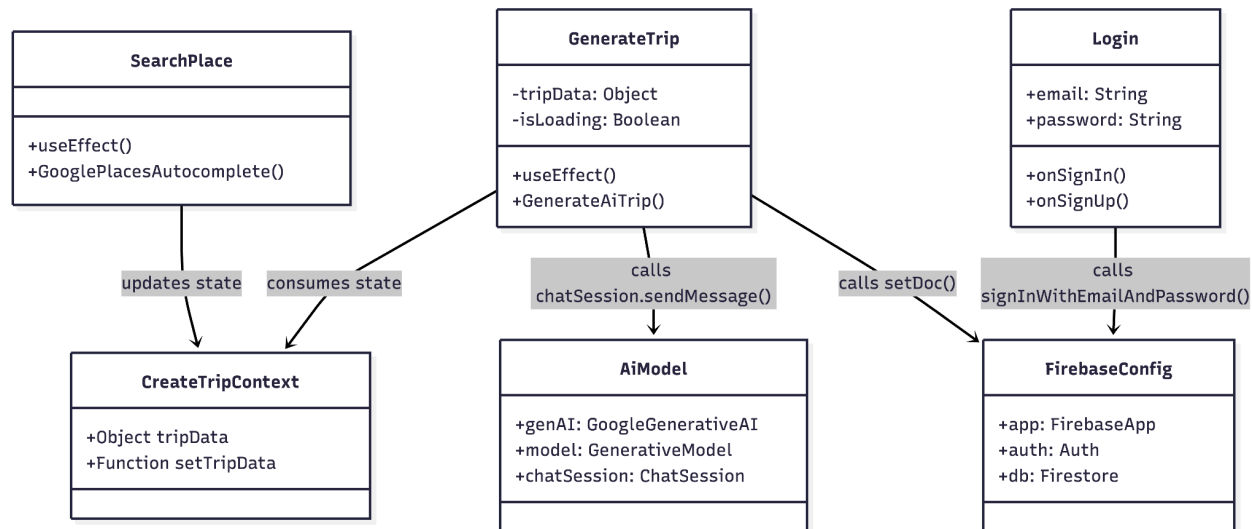
The **context/** module manages shared data, like trip information, and allows multiple screens and UI components to access and update that data without passing it manually between files. The **constants/** components provide shared values like colors and predefined data that are used by components and screens to maintain consistent design and behaviour. Additionally, the **assets/** components provide the images and fonts used by the UI. Finally the **scripts/** and **root configuration files** support the development and execution of the project by managing dependencies, build settings and utility tasks needed to run the application.

c. Include at least one UML diagram that clearly shows high-level structure of the system.



3. Design Details (7 points)

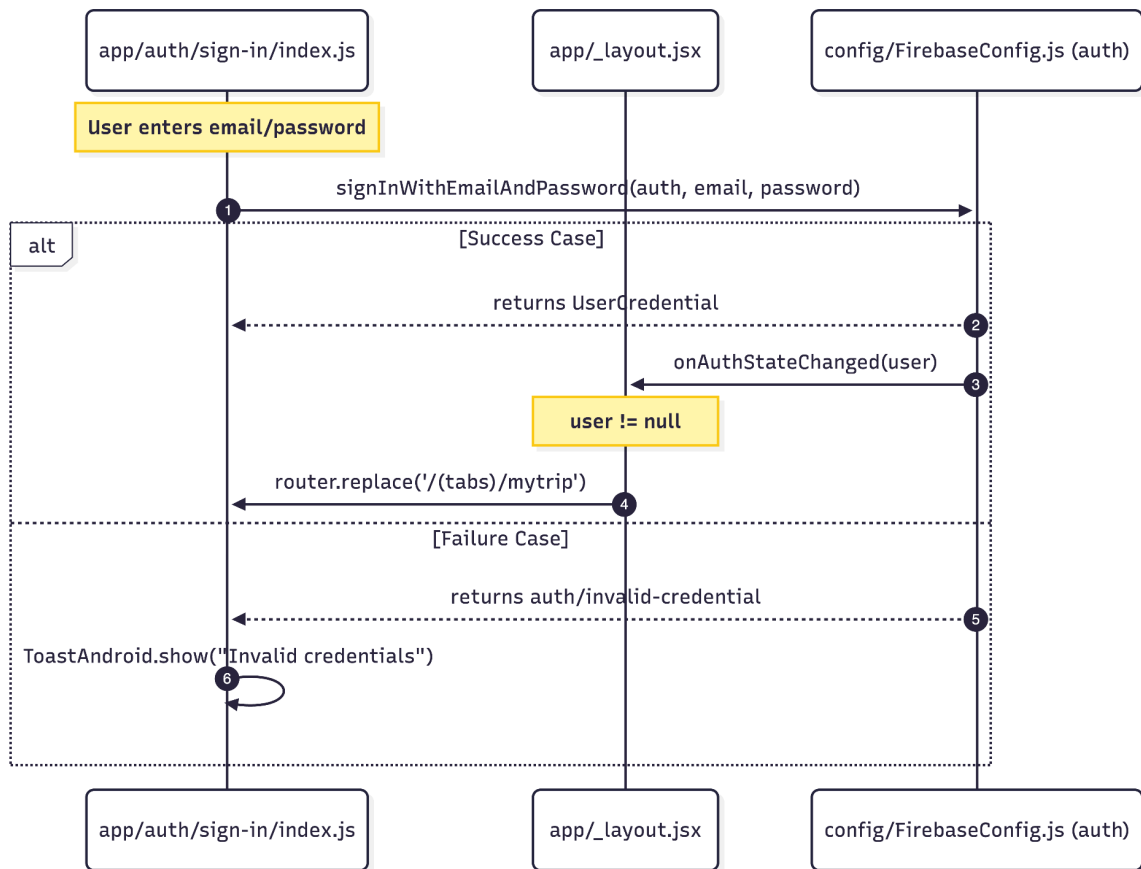
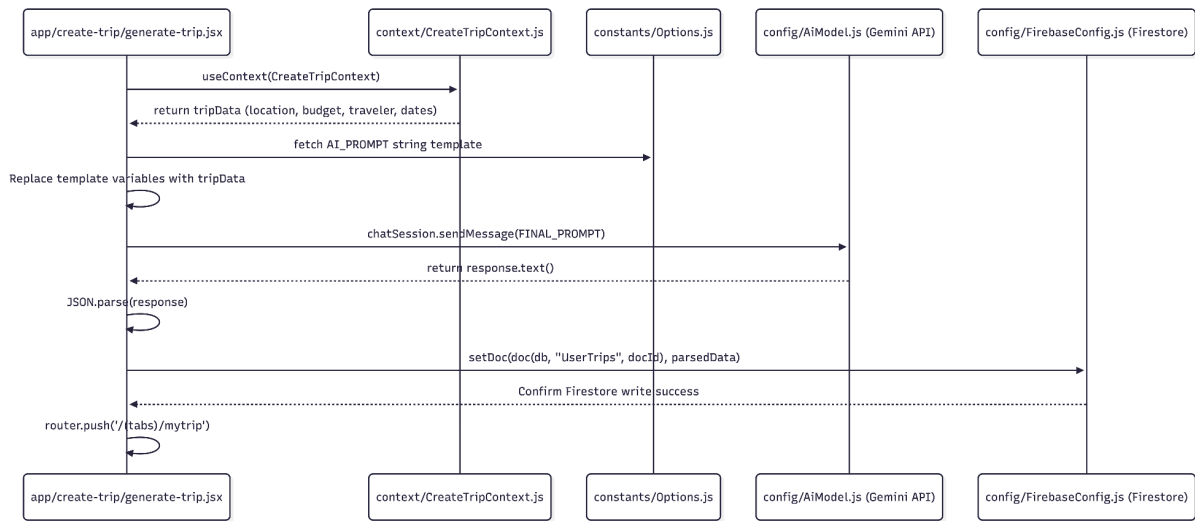
a. Include class level design (i.e., class diagrams) and be as detailed as you can.

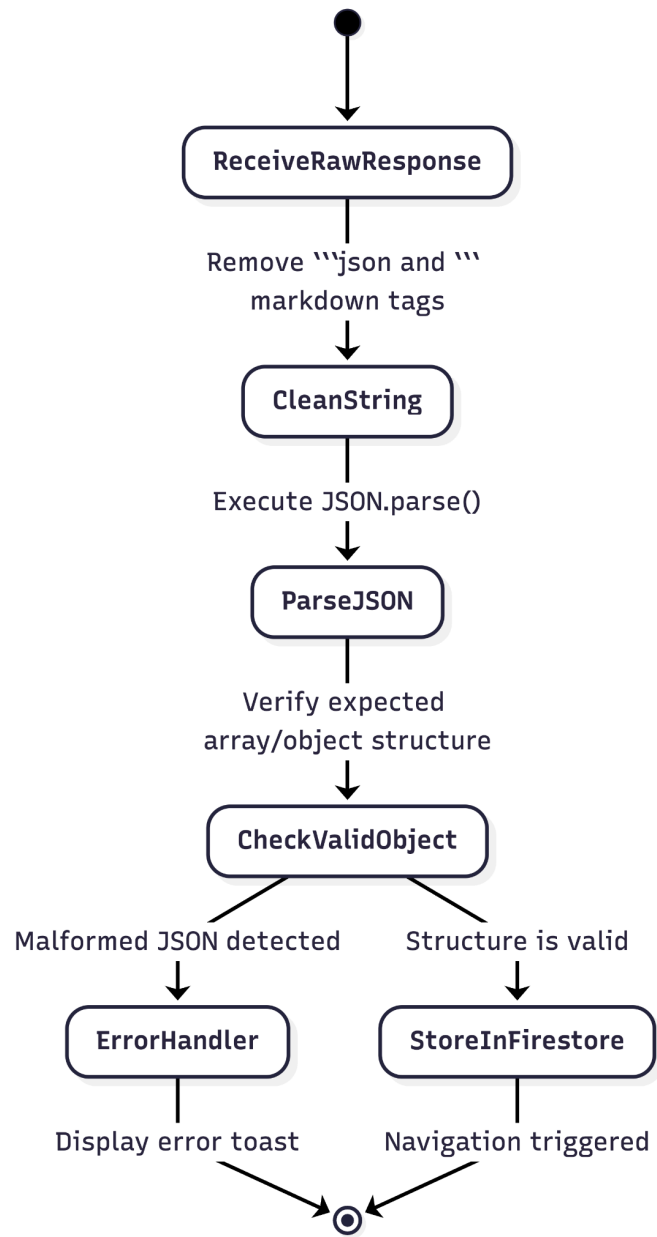
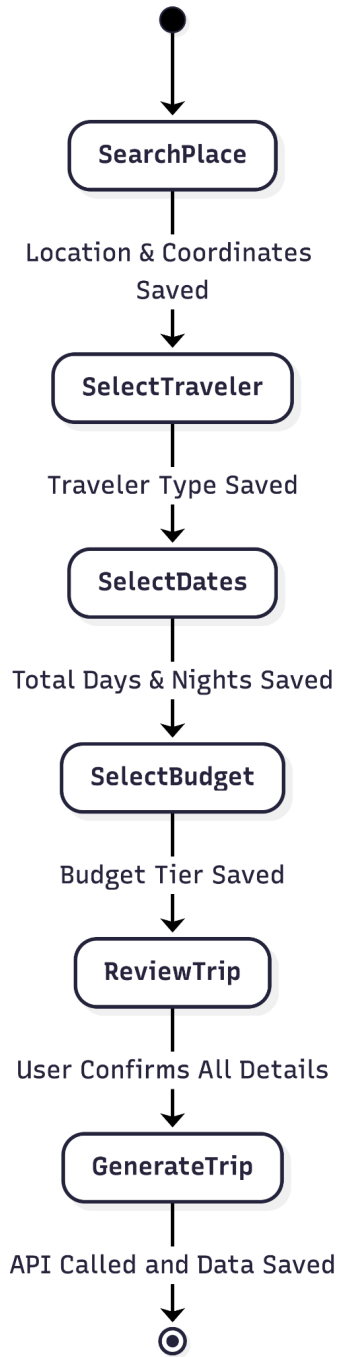


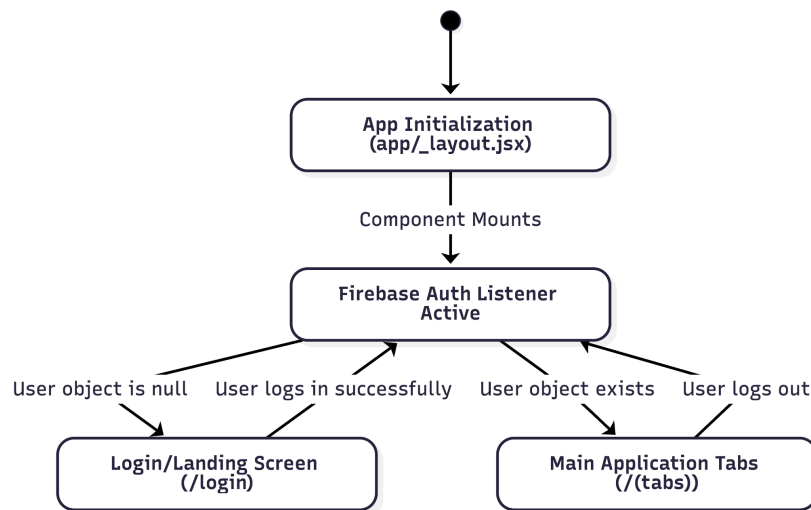
b. Describe the classes and interactions between the classes.

- **CreateTripContext:** This module in context/CreateTripContext.js acts as the main storage for a trip while a user is still picking their options. It holds the userTripData object and provides the setUserTripData function so every screen in the flow can update the destination, budget, or traveler count.
- **GenerateTrip:** This component in app/create-trip/generate-trip.jsx handles the logic of turning user choices into a final plan. It reads from the context, sends the formatted data to the AI, and manages the process of parsing the result and saving it to the database.
- **AiModel:** Located in config/AiModel.js, this file sets up the connection to the Gemini API. It configures the AiChatSession with the specific parameters and safety settings required to get a structured JSON response from the model.
- **FirebaseConfig:** This configuration file in config/FirebaseConfig.js initializes the Firebase app. It exports the auth and db instances used by the login screens and the trip generation logic to communicate with the backend.
- **SearchPlace & Login:** These are input-focused classes. SearchPlace uses the Google Places API to update location data in the context, while Login (in app/auth/sign-in/index.js) manages user credentials through the Firebase authentication service.

c. Include sequence diagrams for different activities and/or activity diagrams and/or state diagrams (at least five UML diagrams).







d. In your descriptions and/or diagrams, point to the source file and line numbers if applicable, so that the grader can have a look.

1. AI Trip Generation (Diagram 1)

a. Constructing the Prompt: In `app/create-trip/generate-trip.jsx` at Line 24, the component pulls the `AI_PROMPT` template from `constants/Options.js` and injects the user's trip preferences.

b. Executing the API Call: The actual request to the Gemini API happens on Line 55 of `app/create-trip/generate-trip.jsx` through the `AiChatSession.sendMessage` method.

c. Saving to the Database: Once the itinerary is received, it is persisted to the "UserTrips" collection in Firestore on Line 73 using the `setDoc()` method.

2. Authentication Logic (Diagram 2)

a. The Sign-In Call: The interaction with Firebase Auth is located on Line 42 of `app/auth/sign-in/index.js`, where `signInWithEmailAndPassword` is invoked.

b. Error Handling: The logic for invalid credentials (the else path in the sequence diagram) is handled by the catch block on Lines 45-50, which triggers a `ToastAndroid` notification.

3. Multi-Step Form State (Diagram 3)

a. Updating Global State: Each input screen in the `app/create-trip/` directory (such as `select-budget.jsx`) calls `setUserTripData` from `context/CreateTripContext.js` to update the central store.

b. Navigation Flow: The sequential movement between screens is managed by `router.push()` calls found at the conclusion of each selection function within those specific screen files.

4. Cleaning the AI Response (Diagram 4)

a. Scrubbing Markdown: Gemini often returns the itinerary wrapped in markdown code blocks; Lines 60-61 of `app/create-trip/generate-trip.jsx` use `.replace()` to strip out the ````json` and ````` tags.

b. Parsing the String: The sanitized string is converted into a standard JavaScript object on Line 64 using `JSON.parse()`.

5. Navigation Guard (Diagram 5)

a. Setting the Listener: The `onAuthStateChanged` observer is initialized within a `useEffect` hook on Line 17 of `app/_layout.jsx`.

b. The Redirect Check: The specific logic that verifies if a user is logged in—and redirects them to the login screen if they are not—is implemented on Line 20 of

app/_layout.jsx.

4. Design Patterns (4 points)

- a. Provide textual description of the patterns along with their diagrams (at least five patterns).**
- b. In your descriptions and/or diagrams, point to the source file and line numbers if applicable, so that the grader can have a look.**

5. Design Issues (1 point)

- a. Identify one functional design issue and one non-functional design issue.**
- b. Each design issue requires a descriptive title, the solution option chosen by the project, and the alternative solution that you can think of which will either degrade/improve the design.**

Submission

Please make sure to include team number and the names of all your team members. You are expected to follow the organization of the design documentation with a right information in the right section and appropriate pointers to the project repository content in your descriptions and/or diagrams.

A homework assignment will be created on Gradescope (only one submission per group). You are expected to submit a document in PDF format.

Violation of the above instructions will result in a reduction in score.

Open Source Project links:

- <https://github.com/zahraaMeky/ai-travel-planner-app>
- <https://github.com/barika001/ai-trip-planner>